LESSON PLAN

ROCKET ENGINE (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of Prepregs
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of Prepregs. Application of prepregs in aerospace industry. Advantages and disadvantages of prepress. Storage of Prepregs.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of prepreg storage conditions and handling safety precautions; Instructor supervision required for safety compliance.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of a composite material?
(b) Why are composite materials preferred over conventional metals in aerospace?
(c) Can you define the term 'prepreg' based on your prior knowledge?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and composition of prepregs • Development history of composite materials • Functions of fibers and resin systems • Fabric weaves and selection criteria • Matrix selection parameters	L & Ds	PPT, IB	15 min
(b)	• Applications in aerospace industry • Advantages of prepreg technology • Disadvantages and limitations • Common defects in composite structures • Strength flowcharts and performance metrics	L & Ds	PPT, PDF	20 min
(c)	• Storage requirements for prepregs • Temperature and humidity control • Shelf life management • Handling and cutting procedures • Safety precautions during storage	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the critical difference between wet layup and prepreg manufacturing methods.
(b) List three specific advantages of using prepregs in aerospace structural applications.
(c) Describe the storage conditions required to maintain prepreg integrity.

12. Home Assignment and Follow Up	Prepare a brief report comparing the stress-strain characteristics of composite prepregs versus conventional metals, referencing the data to be covered in LP 06.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Comparison of composite with conventional metal w.r.t stress-strain graphical diagram.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min